

HAMEENLINNA KATINEN, Finland

WMO: 027540

Lat: 60.9992N

Lon: 24.4917E

Elev: 279

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -3.5	(d) -13.7	(e) 2.6	(f) -8.7	(g) -7.8	(h) 3.7	(i) -2.9	(j) 22.5	(k) 34.4	(l) 20.4	(m) 34.4	(n) 2.5	(o) 100	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	81.7	65.9	78.1	63.8	74.8	62.4	68.9	77.0	66.6	74.8	64.4	72.0	8.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	97.1	72.8	63.5	88.6	70.4	61.2	81.4	67.9	33.1	76.8	31.3	74.3	29.6	72.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.0	(b) 17.8	(c) 15.9	(e) -14.1	(f) 84.8	(g) 6.3	(h) 4.3	(i) -18.6	(j) 87.9	(k) -22.2	(l) 90.4	(m) -25.7	(n) 92.9	(o) -30.3	(p) 96.0		
(4) 20.0	(5) 17.8	(5) 15.9	(5) -12.8	(5) 70.2	(5) 8.1	(5) 3.2	(5) -18.6	(5) 72.5	(5) -23.3	(5) 74.3	(5) -27.9	(5) 76.1	(5) -33.8	(5) 78.4		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	19.3	23.4	29.0	39.8	51.8	58.0	64.2	60.7	53.0	41.0	34.9	27.3
(7)		DBStd	16.78	11.91	11.15	8.16	6.38	7.62	6.28	5.87	5.33	5.48	6.39	7.39	10.03
(8)		HDD50	4215	953	746	652	311	70	8	1	33	284	454	703	
(9)		HDD65	8514	1418	1166	1117	755	413	225	89	154	361	744	904	1168
(10)		CDD50	1281	0	0	0	7	125	248	439	333	123	6	0	0
(11)		CDD65	102	0	0	0	0	3	14	63	22	0	0	0	0
(12)		CDH74	873	0	0	0	0	72	157	502	142	0	0	0	0
(13)		CDH80	159	0	0	0	0	6	21	105	27	0	0	0	0
(14)	Wind	WSAvg	7.7	7.1	7.2	8.1	7.7	7.9	8.0	7.4	7.1	7.3	7.8	8.4	8.4
(15)	Precipitation	PrecAvg	25.1	2.1	1.6	1.3	1.3	1.5	2.5	2.8	3.0	2.0	2.6	2.3	2.1
(16)		PrecMax	30.5	4.0	3.3	2.7	2.7	2.8	4.4	7.4	6.4	5.8	6.6	5.6	4.6
(17)		PrecMin	21.0	0.3	0.1	0.2	0.2	0.4	1.0	0.1	0.6	0.4	0.6	0.2	0.2
(18)		PrecStd	2.9	0.9	0.9	0.6	0.7	0.7	0.9	1.6	1.3	1.3	1.3	1.1	1.0
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	39.5	42.2	50.6	68.1	80.7	83.0	86.5	83.7	70.4	58.0	50.4	46.6	
(20)		0.4% MCWB	38.1	37.5	41.5	51.8	60.1	64.1	67.9	67.3	60.9	53.4	48.4	45.5	
(21)		2% DB	37.9	38.7	45.1	61.9	75.7	78.6	83.0	77.5	66.9	54.5	48.2	42.2	
(22)		2% MCWB	36.7	36.4	38.6	48.0	58.5	61.6	67.8	64.6	59.4	51.4	46.9	40.5	
(23)		5% DB	35.8	36.7	41.7	57.0	71.4	74.7	79.7	74.0	64.0	52.7	45.6	40.1	
(24)		5% MCWB	35.0	35.2	37.0	45.8	56.9	60.2	65.9	62.5	57.2	50.5	44.3	38.8	
(25)		10% DB	33.8	35.1	38.8	52.2	67.0	70.9	76.1	70.9	61.7	50.7	43.6	38.0	
(26)		10% MCWB	33.0	34.1	35.4	42.6	54.2	58.6	64.9	61.3	56.6	48.5	42.5	36.8	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	38.6	38.4	42.7	52.8	63.6	68.4	71.6	70.9	62.1	54.9	49.1	45.6	
(28)		0.4% MCDB	39.0	40.6	47.3	64.4	73.3	75.8	80.9	79.0	67.0	57.2	50.3	46.6	
(29)		2% WB	36.9	36.8	39.6	49.6	60.9	64.5	69.8	66.9	60.6	52.6	46.6	41.0	
(30)		2% MCDB	37.7	38.3	43.6	60.2	70.9	75.1	78.0	73.9	65.1	53.8	47.9	41.8	
(31)		5% WB	35.2	35.5	37.8	46.6	58.4	62.0	68.1	64.6	59.1	51.0	44.5	39.0	
(32)		5% MCDB	35.9	36.7	40.9	54.6	68.3	71.8	76.7	71.3	62.6	52.3	45.6	40.0	
(33)		10% WB	33.2	34.1	35.9	44.0	55.7	59.9	66.1	62.6	57.4	48.6	42.7	37.0	
(34)		10% MCDB	33.9	35.0	38.3	50.8	65.3	68.2	74.5	68.7	60.5	50.3	43.4	38.0	
(35)	Mean Daily Temperature Range	MDBR	9.6	9.4	13.1	16.5	18.5	16.4	15.7	15.2	12.9	10.2	6.4	7.3	
(36)		5% DB	6.6	8.4	15.0	25.6	26.2	23.2	20.9	20.6	16.8	12.3	7.6	7.6	
(37)		MCWBR	6.4	7.2	9.9	13.8	12.1	9.9	8.3	9.4	9.7	9.2	7.1	7.3	
(38)		5% WB	6.3	7.5	13.4	22.0	21.9	20.1	18.0	17.4	14.3	10.3	7.3	7.7	
(39)		MCWBR	6.2	6.7	9.5	12.6	11.4	9.5	8.4	8.7	9.0	8.9	7.0	7.5	
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.299	0.326	0.357	0.355	0.351	0.375	0.363	0.338	0.324	0.284	0.371	
(41)		taud	2.095	2.165	2.245	2.368	2.433	2.464	2.417	2.443	2.509	2.405	2.194	2.619	
(42)		Ebn at Noon	146	212	250	266	276	279	268	261	249	207	137	100	
(43)		Edh at Noon	13	23	29	31	32	31	32	29	23	18	12	8	
(44)	All-Sky Solar Radiation	RadAvg	75	254	723	1192	1609	1679	1625	1256	749	323	89	42	
(45)		RadStd	14	56	92	124	176	133	158	139	99	56	17	6	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (61 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	-261	N/A	N/A	N/A	(47)

Nomenclature: See separate page